Aryan Goel

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MS Data Science candidate focused on causal inference, experimentation, and applied ML — turning messy data into decisions a business can actually act on.

EDUCATION

University of Southern California — Los Angeles, CA
Master of Science, Computer Science (Data Science Specialization) — *Aug. 2025 – May 2027 (Expected)*
***Relevant Coursework**: Machine Learning for Data Science, Applied Statistical Inference, Deep Learning, Algorithms, Database Systems (indexing, storage engines, query optimization), Foundations of Data Management (Hadoop, Spark, distributed SQL), NLP*

Manipal University Jaipur — Jaipur, India
Bachelor of Technology, Computer & Communication Engineering — GPA: 9.15/10 — *July 2021 – May 2025*

TECHNICAL SKILLS

Languages: Python, SQL, R (basic)
Statistics & Experimentation: Hypothesis Testing, ANOVA, A/B Testing, Power Analysis, Confounding/Simpson's Paradox Analysis, Causal Reasoning, Uplift Modeling (S/T/X-learners), Probability Calibration
ML & Modeling: Scikit-learn, PyTorch (CNNs), Logistic Regression, Random Forest, Gradient Boosting, Class Imbalance (SMOTE), Feature Engineering, XGBoost, Isolation Forest/LOF, Minimax/Alpha-Beta Search, Reinforcement Learning (prototype)
Explainability & Evaluation: SHAP, Grad-CAM, Precision/Recall/ROC-AUC, Sensitivity-Weighted Metrics for Imbalanced Data
NLP & GenAI: Hugging Face Transformers, Text Summarization (ROUGE), Tokenization & Embeddings, Multi-Agent Orchestration (AutoGen)
Data & Databases: Pandas, NumPy, SQL (joins, window functions, query optimization), PostgreSQL, DuckDB, Indexing & Storage Fundamentals, MySQL, ClickHouse, PySpark
Tools & Cloud: Matplotlib, Seaborn, Git, AWS EC2, GCP (Vertex AI), Streamlit

EXPERIENCE

Indian Institute of Remote Sensing (ISRO) — Dehradun, India
Research Intern — *Feb. 2025 – May 2025*

- Engineered a cascaded CNN architecture (CED-Net) for crop/weed segmentation on RGB drone imagery, outperforming a baseline U-Net through residual blocks and attention mechanisms
- Built the full annotation-to-evaluation pipeline (LabelMe → pixel masks → training/eval on **IoU and Dice score**), benchmarking multiple architecture variants

Indian Institute of Technology (IIT) Roorkee — Roorkee, India
Research Summer Engineer — *May 2024 – July 2024*

- Devised a Physics-Informed Gaussian Process Regression model to forecast flood events, embedding hydrological constraints directly into the model for improved reliability over a standard GPR baseline
- Executed **uncertainty quantification** via Monte Carlo and Multi-Level Monte Carlo simulations for confidence-bounded risk predictions

Indian Institute of Technology (IIT) Roorkee — Roorkee, India
Software Developer — *June 2023 – July 2023*

- Managed and optimized geospatial data layers in **GeoServer** for a production WebGIS application; performed data classification with QGIS
- Redefined the rendering pipeline and resolved performance bugs to reduce load time, while shipping new data-classification features

PROJECTS

Telecom Churn & Retention Intelligence | *Python, scikit-learn, XGBoost, SHAP, DuckDB*

- Engineered a leakage-safe churn pipeline on synthetic data with a built-in Simpson's paradox: priority_support looked 6.5pp worse on raw churn yet lowered churn in every risk quartile once stratified (self-selection by high-risk customers).
- Compared logistic regression and XGBoost with nested CV and in-pipeline SMOTE (PR-AUC 0.593 vs. 0.587), isotonic-calibrated the probabilities (SMOTE inflated them ~5pp), and set a 0.18 cost-optimal threshold at a 5:1 miss-to-offer cost ratio, cutting expected cost by more than half.
- Estimated heterogeneous effects with S/T/X-learner uplift models (Qini/AUUC) and a simulated A/B test; re-ran the pipeline on real IBM Telco data (ROC-AUC 0.840), where the paradox did not replicate and was reported as such.

ChainSignal: Crypto Market & On-Chain Anomaly Detection | *Python, PySpark, ClickHouse, scikit-learn*

- Built a ClickHouse + PySpark pipeline over 1,000 coins × 365 days and 1.64M on-chain transfers, with no-lookahead features, robust z-scores, and exclusion of 83 feed-error coins.
- With no labels, compared a robust z-score, Isolation Forest, and LOF using 887 planted anomalies and a backtest on documented events: Isolation Forest recovered 43% of 12σ combination anomalies vs 11% for z-score at a 1% alert budget; the Kelp DAO hack was missed and reported as a limitation.
- Flagged Celer Network's $1.5B volume on an $18M market cap with flat price as a wash-trading candidate (confirmed in two aggregators).

Little-Go: Adversarial Search Game-Playing AI Agent | *Python, Minimax/Alpha-Beta Pruning, Reinforcement Learning*

- Built a competitive Go-playing agent (5×5 board, USC CSCI-561) using minimax search with alpha-beta pruning and a custom heuristic evaluator, winning 60/60 games in the course's automated grading tournament.
- Ran a 200–400 game tournament vs. the prior version, measuring a 63.5% win rate (95% CI) to confirm the gain wasn't noise.
- Built a 61-test mutation-tested suite (12 planted bugs) for real regression coverage; a separate RL pipeline improved endgame win-margin prediction by up to 5.7 points, not yet in the submitted agent.

SolarWatch GB: Weather-Adjusted Solar Anomaly Detection | *Python, MySQL, scikit-learn*

- Fit per-region baselines (generation vs. irradiance, through-origin least squares; median R^2 0.81) across 323 UK regions × 60 days (465K hourly rows) and flagged regions below 70% of expected output (4.9% of region-days).
- Validated out of sample (R^2 0.76) and against PV_Live's national total (98.9% coverage), since no ground truth exists.

Book Recommender — item-based collaborative filtering (cosine kNN on Book-Crossing) with a same-author content row, built on a course codebase *(Python, scikit-learn, Streamlit)*

ACHIEVEMENTS

- Second Runner-Up, UCLA Glitch × Google DeepMind Hackathon — built BrandSync, then developed it fully after the Google team's positive feedback
- 1st Place, Inter-College Hackathon (Mental Health track) — built a Figma-prototyped app with an integrated chatbot
- Dean's List for Academic Excellence, Manipal University Jaipur (awarded twice)